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Correlation-Based Approach to Color Image Compression

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Abstract

Most coding techniques for color image compression employ a de-correlation approach - the RGB primaries are transformed into a de-correlated color space, such as YUV or YCbCr, then the de-correlated color components are encoded separately. Examples of this approach are the JPEG and JPEG2000 image compression standards. A different method, of a Correlation Based Approach (CBA) is presented in this paper. Instead of de-correlating the color primaries, we employ the existing inter-color correlation to approximate two of the components as a parametric function of the third one, called the base component. We then propose to encode the parameters of the approximation function and part of the approximation errors. We use the DCT (Discrete Cosine Transform) block transform to enhance the algorithm's performance. Thus the approximation of two of the color components based on the third color is performed for each DCT subband separately. We use the Rate-Distortion theory of subband transform coders to optimize the algorithm's bits allocation for each subband and to find the optimal color components transform to be applied prior to coding. This pre-processing stage is similar to the use of the RGB to YUV transform in JPEG and may further enhance the algorithm's performance. We introduce and compare two versions of the new algorithm and show that by using a Laplacian probability model for the DCT coefficients as well as down-sampling the subordinate colors, the compression results are further improved. Simulation results are provided showing that the new CBA algorithms are superior to presently available algorithms based on the common de-correlation approach, such as JPEG.

Key words: Color image compression, Inter-color correlation, Correlation Based Approach, DCT block transform, Rate-Distortion model, Color components transform, Optimal rates allocation

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